Cost of Size

What size costs on a Kalshi event contract, published beside the one-contract price Pyth already carries.

29	19 of 24	29%	5.9×
hedge blocks in 17 markets, June 4–November 6; median 250k contracts	quoted blocks printed outside the bid–ask, median 3¢ through	of 56 books fill 250k on both sides (57% fill one side)	a 250k block vs the median book's depth within 5¢ of mid

What the product is

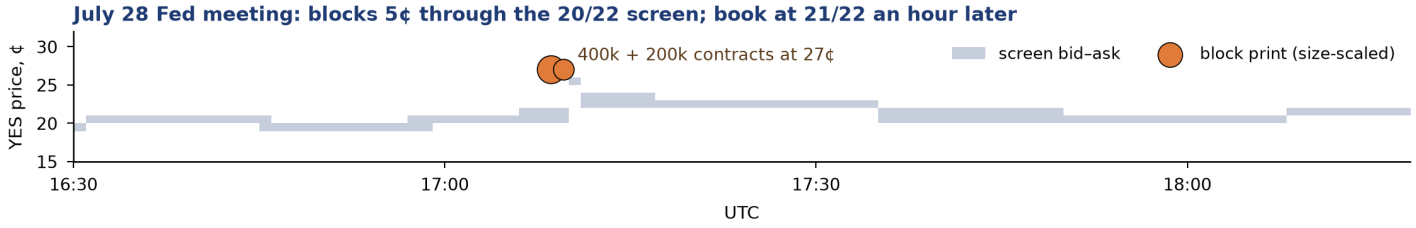
The screen answers one contract. Hedgers trade 250,000. Cost of Size publishes, per contract, the cost in cents to fill 25k, 100k, 250k, and 1M on each side, walked from the live book; a flag when the book cannot fill; and a band each new block print is checked against. That is phase 1, from the public book. Phase 2 adds the size quotes dealers pass, so the same curve covers block and OTC interest the tape never shows. The screen stays the price. Execution stays with Kalshi.

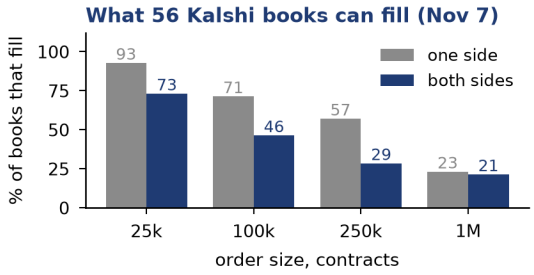
Why it's needed

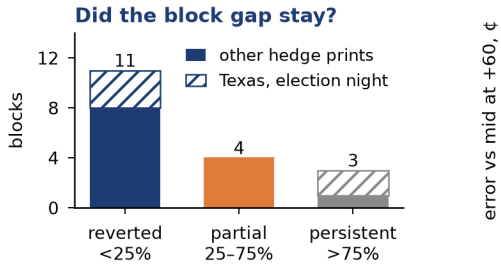
The gap. The only public number for a Kalshi contract is the one-contract screen. Nothing says what 250,000 costs. The size prints the tape flags, blocks, trade a median 3¢ through that screen and mostly fade within the hour. The largest hedges never reach the venue at all. Anyone sizing, quoting, or marking a large position is pricing it blind.

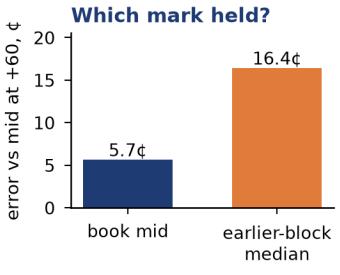
Who uses it, and what it fixes

- **Hedgers sizing a trade.** The 29 blocks are hedges on elections (11), the economy and the Fed (6), and commodities (5). Today they learn the cost at the print: on July 28, 600k Fed contracts went at 27¢ against a 20/22 screen. The curve tells them first: at 250k, 29% of books fill both sides, and where they do, a side costs a median 8.2¢.
- **Block desks and market makers quoting size.** 19 of 24 quoted blocks printed outside the bid–ask, 4 on the touch, and 1 inside. The concession is agreed privately. A public cost curve turns it into a number both sides quote against and anyone can check after.
- **Risk desks, and protocols marking with Pyth's Kalshi price.** The screen overstates what a 250k position exits for: that size is 5.9× the median book's depth within 5¢. Marking to block prints is worse. On the same 9 prints, the earlier-block median missed the mid an hour later by 16.4¢ and the book by 5.7¢. Of 18 gaps an hour later, 11 reverted (under 25% left), 4 were partial (25–75%), and 3 were persistent (over 75%). 5 of the 18 are Texas election-night prints.
- **Funds that leave the exchange entirely.** In May 2026 Galaxy and Arca did a $10 million swap on whether the CLARITY Act passes Congress. It was executed OTC, through Galaxy, and never sent to Kalshi or Polymarket. Arca said a fund of its size cannot get that done on the screen. The contract was marked to Kalshi, where it was listed at 37¢, so the one-contract price was the reference for a trade the book never saw.









How Pyth integrates it

- **Phase 1 is the public feed.** Kalshi's order book and the trade feed that flags blocks of 25,000+. No new data partner. Publish the cost at 25k / 100k / 250k / 1M, the can't-fill flag, and the band beside the existing Kalshi price, and score each block when it prints. This run has one November 7 book, and 1 of the 29 blocks is in it, so storing the book continuously comes first. A block can be reported up to 15 minutes after agreement, so the check uses the book at agreement time.
- **Phase 2 is the dealers.** The public tape is not the block market: Galaxy and Arca's swap never reached Kalshi. Dealers pass the size quotes they show off the screen, and those quotes extend the curve to the block and OTC interest the book never records.

Kalshi public API, June 4–November 6: 52 blocks, 52.2M contracts; sports and parlays out. Scan is a lower bound; on-book RFQs are not flagged. Book: November 7, 56 markets. Galaxy–Arca from Galaxy (June 2, 2026) and Arca; not in this tape.

Benchmark funding rate

Crypto has no reference rate for the cost of leverage. Pyth should publish one: an hourly, market-wide funding rate with a confidence band.

48% of days
ETH funding on Hyperliquid and Binance more than 5 pp apart (annualised), Nov 2025–Oct 2026

$2,284
median gap on a $1M 30-day ETH funding swap, Hyperliquid vs the composite ($3,939 at p95)

267% vs 11.9%
Lighter SOL funding vs the composite in the same hour, 25 Aug 2026

0
onchain funding rates built from more than one venue (Boros, Fandy, Pindar, Clamp each settle on one)

What the product is

An hourly, market-wide funding rate across Binance, OKX, Gate, Hyperliquid and Lighter: open-interest weighted, capped at 30% per venue, so no venue sets the rate alone (Binance holds 43–55% of open interest), the same cap CF Benchmarks uses; a median alongside it; and a confidence band measuring how far venues disagree, published onchain like a Pyth price. The venue feeds it is built from are published too, upgraded from today's single publisher to several. Products and settlement stay with the protocol.

Why it's needed

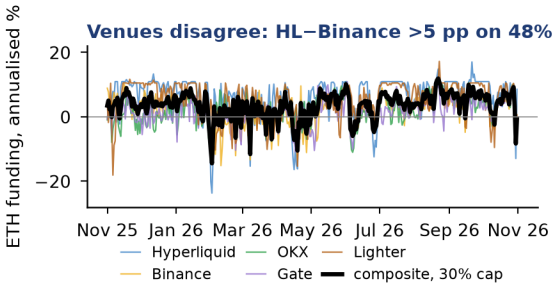
Crypto has no reference rate for the cost of leverage. Every funding product settles on one venue's print, as if every loan were priced off one bank's lending rate. Funding is already traded (Boros has passed $20B of cumulative notional across 242 markets, four of them cross-venue spreads and the rest each on one venue), venues disagree (ETH on Hyperliquid and Binance more than 5 pp apart on 48% of days), and single venues spike (Lighter SOL 267% while the composite stayed at 11.9%). A market-wide rate is what fixed-rate products, structured yields and lending rates can be written against, and whoever publishes it becomes the reference.

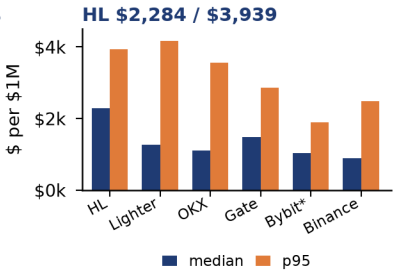
Provider	What it does	Market-wide?	Frequency	DEX venues	Onchain
Chainlink Data Streams / Chaos Labs / Pendle oracle	Verify one venue's print for Boros	No	Per print	Yes	Yes
Chainlink DataLink	Single-source transport, no source validation	No	Per source	n/a	Yes
Coinglass	OI-weighted funding, analytics	Yes	Intraday	Yes*	No
CF Benchmarks market-wide index	Regulated benchmark, 30% cap	Yes	Once a day	No	No
Pyth (proposed)	Reference rate + confidence band + venue feeds	Yes	Hourly	Yes	Yes

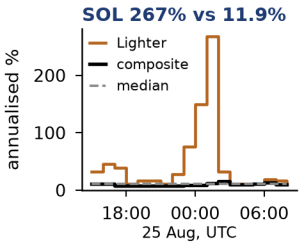
Who uses it, and what it fixes

- **Products that need a reference rate.** Fixed-rate funding, structured yields ('funding plus 2%') and leverage lending need a rate no single venue can move; today none exists onchain, so each product picks one venue's print.
- **Funding-swap protocols (Boros and others).** Their oracles check that one venue's print was reported correctly; none publishes the market rate. The composite lets them list market-wide funding beside the single-venue and spread markets already traded, and the confidence band flags a print like Lighter's 267% instead of averaging it in.
- **Holders of funding across venues.** A book that collects funding on several exchanges is not hedged by one venue's print. On ETH the leftover basis was $2,284 median / $3,939 p95 per $1M per month, Hyperliquid against the composite.
- **Risk teams holding funding-dependent collateral.** The band shows when funding is fragmenting or turning negative across venues, an early signal for basis positions and funding-yield collateral.

Signs of demand: CF Benchmarks launched a market-wide funding index in July 2026, and Boros launched cross-venue spread markets in October 2026.







How Pyth integrates it

- **Phase 1.** Publish the hourly reference rate and confidence band on the funding feeds Pyth already runs, and add publishers to each venue feed (11 live feeds require only one publisher). Venues can publish their own funding directly, as they publish prices today: Binance, OKX and Hyperliquid are on pyth.network/publishers, and Bybit and Gate.io publish too.
- **Phase 2.** Equity-perp funding is a good opportunity as equity perps pick up.

Sources and limits. 12 months Nov 2025–Oct 2026; Binance via data.binance.vision; Bybit* from the Boros historical archive (Bybit API geo-blocked; archive matched primary data exactly for 4 venues); OI weights: Nov 2026 snapshot, held fixed. Composite capped at 30% per venue; a 20% cap equal-weights the five venues and took the SOL composite to 61% at the spike, so 30% is used (a6). 8h prints spread over the hour; part of a venue gap is basis a hedger wants. *CoinGlass publishes Hyperliquid futures (coinglass.com/exchanges/Hyperliquid); its OI-weighted endpoint does not list constituents. Sources: docs.cfbenchmarks.com; pyth.network/publishers; legacy.pyth.network/publishers; pyth.dourolabs.app; docs.pendle.finance/boros-docs; historical-data.boros.finance; cryptobriefing.com; docs.chain.link/datalink.md; docs.coinglass.com; fandy.finance; pindar.finance; clamphyperevm.com.